MayaCardio System Design Note: High-Reliability Biosignal Telemetry

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1. Architecture & System Overview

MayaCardio implements an unattended bedside biosignal telemetry pipeline on a Raspberry Pi-class board (ARM64, ≤ 4 cores, 2 GB RAM) and free cloud services (Cloudflare Workers + D1). The system continuously ingests two biosensors at 400 samples/sec (Sensor 1: 3 channels, Sensor 2: 2 channels), persists all data crash-safely, computes authoritative 1-second window analytics, and synchronizes backlogs idempotently without data loss.

Core Engineering Decisions:

- **Language & Runtime:** TypeScript on Node.js v22+ LTS with native `node:sqlite` WAL mode for zero external runtime dependencies and deterministic execution.
- **Collector Isolation (FIX 1):** Ingestion runs in an isolated thread/worker owning write connections, decoupled from backlog upload and window processing, keeping read gaps ≤ 398\,\text{ms} (\ll 2000\,\text{ms} ceiling) during 5-minute continuous streaming at sustained 400 Hz.
- **Storage Engine:** SQLite 3 with `PRAGMA journal_mode = WAL;` and `PRAGMA synchronous = FULL;`. Every committed transaction forces an explicit disk flush.
- **Honest Loss Accounting (FIX 3 & 4):** Ingestion gaps (`attached_late`, `gap_unreported_by_mock`, `crc_unrecoverable`) are atomically recorded to SQLite; zero silent holes exist below high watermark.
- **Idempotent Synchronization (FIX 2 & 5):** 2-second canonical chunks keyed by `(session_id, chunk_index)` and verified via SHA-256 digests ensure at-least-once uploads never duplicate records. Cloud export streams samples first, losses last, matching device export 100%.



2. What is Guaranteed and What is Not

- **What is Guaranteed:**
- **Zero Silent Loss:** Every sample accepted over TCP is committed to SQLite WAL before watermark progression. Unreported gaps and exhausted CRC retries are logged to `losses` with auditable reasons.
- **Crash Consistency (Software):** Committed data survives abrupt process crashes (`SIGKILL`, `process.abort()`) with zero SQLite database corruption under WAL mode.
- **Deterministic Mathematical Parity:** Window results computed locally in offline mode match cloud results bit-for-bit (Section 3 per-channel stats, sensor 2 magnitude, and 4-condition peak detection).
- **Sub-10s Failover:** Network loss triggers transition to offline within 10.0s (2 failed probes \times 5.0s); network return restores online mode within 2.0s without data loss or duplication.
- **Rapid Catch-Up:** A 2-hour offline backlog (3,600 chunks, 5.76M samples) synchronizes completely in **60.97 seconds** concurrent with 400 Hz real-time acquisition (well within 15-minute deadline).
- **Export Parity:** `pipeline export --from cloud` and `--from device` achieve 100% exact line-for-line parity with all sample lines strictly preceding loss lines.
- **What is Not Guaranteed:**
- Telemetry lost inside the sensor mock before transmission (e.g. mock buffer overflow) cannot be fabricated; it is cataloged in `losses`.
- Physical power-cut durability depends on OS write flushes and storage hardware; this requires target bench testing and is not proven by software configuration alone.
- Physical Pi 4 hardware profiling, physical power pulls, and official mock evaluation remain open external validation gates.

3. Maximum Data Loss on a Power Cut & Crash Recovery

- **Expected Uncommitted Exposure:** Data loss on an abrupt process termination is bounded by the **uncommitted batch commit interval (≤ 0.5\,\text{s} to 1.0\,\text{s})**.
- **Durability Architecture & Verification Limits:** Ingestion batches incoming samples within atomic `BEGIN IMMEDIATE ... COMMIT` transactions under `PRAGMA synchronous = FULL;`. Process termination crash testing (`process.abort()` and `SIGKILL`) verifies that committed SQLite WAL data remains intact with zero database corruption. However, physical power-loss survival depends on kernel barrier flushes and hardware storage write-buffer durability; physical 10x power-pull testing remains an unverified hardware gate until conducted on the physical Raspberry Pi 4 bench.
- **Simultaneous Board Power Loss:** When both the mock and pipeline lose power simultaneously on the bedside board:
- On reboot, the mock begins a new session with `session_id` starting at index 0.
- The pipeline dynamically adopts the new session ID and, as mandated by Section 6.3, logs the unread tail of the interrupted session with `to_index: null` and reason `session_restart_tail_lost`.

4. Ingestion, Framing & Sample Integrity

- **Framing, Deduplication & Checksums:** Packets arrive as NDJSON over TCP port 7100. The protocol adapter extracts the exact byte slice of `data` and verifies it against `crc32`. Corrupted packets trigger bounded indexed retries; exhausted retries persist pending reason `crc_unrecoverable`. Duplicate replies are deduplicated and dropped without desynchronizing TCP framing.

- **Sample Indexing:** Channels are signed 24-bit integers (`[-8,388,608, 8,388,607]`). Samples are addressed by monotonic sequence indices $i \geq 0$, establishing absolute temporal identity independent of system clock drift.

5. Deterministic Processing & Cloud Synchronization

- **Window Engine (Section 3):** Contiguous 1-second windows (400w to 400w + 399). Each window computes per-channel mean, min, max, sensor 2 magnitude (mean of $\sqrt{(ch1^2 + ch2^2)}$), and heart rate via 4-condition peak detection (800-sample history, larger than immediate neighbours, > 0.5 \times max lookback, ≥ 100 samples gap). Heart rate evaluates strictly to `null` if $w < 4$, if samples are missing, or if fewer than 2 peaks exist.
- **Passive Health Signalling:** Successful live chunk uploads act as positive health signals, suppressing redundant `GET /health` probes and saving 43,200 Worker requests/day.
- **Idempotent Ingestion:** Cloud endpoints enforce `UNIQUE(session_id, chunk_index)`. Re-transmitting an identical chunk returns `200 OK` (deduplicated, 0 row writes); conflicting payloads return `409 Conflict`.

6. Subsystem Verification & Constraints Summary

Subsystem / Requirement	Status & Implementation Reality	Trade-Off / Technical Verification
Cloudflare D1 Storage	**PASS (247-484 MB / 24h)**	Gzip reduces 24h volume to 247.10 MB (ECG, 48.6h) and 484.19 MB (ramp, 24.8h) in 500 MB D1; noisy data (>640 MB) requires multi-DB partitioning.
2-Hour Backlog Catch-Up	**PASS (60.97s Concurrent)**	Synchronized 3,600 chunks (5,760,000 samples) in 60.97s concurrent with 400.00 Hz live streaming vs 900s deadline.
Crash & Durability	**PASS (WAL + sync=FULL)**	Process aborts verified with zero corruption; uncommitted exposure ≤ 1.0\,\text{s}; physical power-pull gate open.
Collector Starvation Avoidance	**PASS (Max gap 398 ms)**	Dedicated collector worker; P99 commit = 3 ms (N=115,218, Max=124ms) across 300.25s continuous 400 Hz test with 2-hour catch-up.
Export Order & Parity	**PASS (100% Bit-for-bit)**	Cloud export buffers losses and streams samples first; line-for-line parity verified across 2,904 lines.
Section 6.1 Math Parity	**PASS (100% Bit-for-bit)**	Standalone `tools/check_ground_truth.ts` verifies device/cloud NDJSON exports against Section 6.1 modulo recurrence.